

TOLUCA INTL, Mexico

WMO: 766753

Lat: 19.337N

Lon: 99.566W

Elev: 2580

StdP: 73.93

Time Zone: -6.00 (MEX)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.0	(c) -0.8	(d) -9.7	(e) 2.3	(f) 18.1	(g) -7.8	(h) 2.7	(i) 16.1	(j) 9.5	(k) 15.7	(l) 8.5	(m) 16.2	(n) 0.4	(o) 140	(p) 0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 5	(b) 15.8	(c) 26.1	(d) 12.1	(e) 25.0	(f) 11.9	(g) 24.0	(h) 11.6	(i) 15.8	(j) 21.0	(k) 15.1	(l) 20.4	(m) 14.8	(n) 20.0	(o) 4.4	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 14.1	(b) 13.9	(c) 17.0	(d) 13.8	(e) 13.5	(f) 16.7	(g) 13.1	(h) 12.9	(i) 16.2	(j) 54.3	(k) 21.0	(l) 52.3	(m) 20.3	(n) 50.8	(o) 20.3	(p) 20.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.4	(b) 7.4	(c) 6.4	(e) -5.2	(f) 28.5	(g) 1.7	(h) 1.2	(i) -6.4	(j) 29.4	(k) -7.4	(l) 30.1	(m) -8.4	(n) 30.8	(o) -9.6	(p) 31.6		
(a) 8.4	(b) 7.4	(c) 6.4	(e) -5.6	(f) 17.1	(g) 1.6	(h) 1.3	(i) -6.8	(j) 18.0	(k) -7.7	(l) 18.8	(m) -8.6	(n) 19.5	(o) -9.8	(p) 20.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	9.8	11.6	13.1	14.9	16.0	15.8	15.1	15.2	15.2	14.0	11.9	10.2		
	DBStd	2.59	1.52	1.65	1.67	1.65	1.55	1.32	1.04	1.01	1.15	1.75	1.87	1.69		
	HDD10.0	50	21	5	1	0	0	0	0	0	0	1	5	17		
	HDD18.3	1741	264	189	163	104	73	76	101	98	94	135	193	251		
	CDD10.0	1354	16	49	97	147	187	174	157	161	156	124	62	25		
	CDD18.3	2	0	0	0	0	2	1	0	0	0	0	0	0		
	CDH23.3	383	2	16	54	124	137	40	2	2	1	2	2	1		
	CDH26.7	25	0	1	1	8	14	2	0	0	0	0	0	0		
(14) Wind	WSAvg	2.0	1.9	2.1	2.2	2.2	2.2	2.1	2.0	2.1	2.1	2.1	1.8	1.7		
(15) Precipitation	PrecAvg	1051	20	18	16	32	83	185	218	205	178	65	22	10		
	PrecMax	1246	93	130	54	78	203	301	381	293	344	116	85	53		
	PrecMin	895	2	1	0	4	6	55	149	135	92	2	2	0		
	PrecStd	109	19	25	14	20	45	51	42	39	58	32	19	12		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	25.1	26.2	27.8	28.2	26.2	23.2	23.2	23.1	23.2	23.2	23.0		
		MCWB	9.9	10.3	11.4	12.8	13.3	12.5	13.5	13.7	13.1	12.3	11.5	10.8		
	2%	DB	21.2	23.2	24.9	26.1	26.2	24.2	22.1	22.1	22.0	22.2	22.0	21.5		
		MCWB	9.4	10.2	11.0	12.1	12.6	12.5	13.7	13.6	13.4	12.2	11.2	10.1		
	5%	DB	20.1	22.1	23.8	24.9	24.9	23.0	21.1	21.1	21.0	21.1	21.0	20.2		
		MCWB	9.3	10.0	10.6	11.8	12.3	12.6	13.5	13.5	13.2	12.2	11.0	9.7		
	10%	DB	18.9	20.9	22.1	23.2	23.2	21.2	19.9	20.0	19.9	20.0	19.8	19.0		
		MCWB	9.1	9.9	10.1	11.1	12.0	13.3	13.5	13.5	13.4	12.2	10.7	9.7		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	13.5	14.2	15.7	16.1	16.3	16.1	16.0	16.1	15.5	14.3	13.2		
		MCDB	19.2	21.4	22.2	23.8	23.0	21.6	20.5	20.3	20.1	20.2	20.3	20.2		
	2%	WB	11.2	12.2	12.9	14.2	15.1	15.2	15.1	15.1	15.1	14.7	13.4	12.1		
		MCDB	18.5	20.4	21.8	23.0	22.6	20.4	19.9	19.7	19.2	19.2	19.2	18.9		
	5%	WB	10.4	11.2	12.0	13.2	14.3	14.8	14.6	14.6	14.6	14.0	12.7	11.1		
		MCDB	17.8	19.3	21.1	21.9	21.6	19.8	19.3	19.2	18.7	18.7	18.3	17.7		
	10%	WB	9.7	10.4	11.0	12.3	13.5	14.2	13.9	14.0	14.0	13.3	11.9	10.2		
		MCDB	17.1	18.4	20.0	20.7	20.3	19.0	18.5	18.5	18.2	17.8	17.5	16.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.1	19.3	19.3	18.2	15.8	12.1	11.3	11.2	10.9	13.6	17.0	18.9		
		MCDBR	21.3	21.4	21.3	20.1	18.3	15.2	13.0	12.8	12.5	15.3	18.6	20.6		
		MCWBR	10.8	10.2	9.5	9.1	7.8	6.5	6.3	6.0	5.8	7.4	9.3	10.5		
		MCWBR	19.3	19.3	19.3	18.3	15.9	12.1	11.6	11.4	10.9	12.6	15.9	18.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.377	0.394	0.401	0.469	0.429	0.481	0.464	0.446	0.349	0.408	0.327	0.367		
		taud	2.202	2.144	2.145	1.982	2.143	1.975	2.045	2.105	2.449	2.190	2.445	2.231		
		Ebn at Noon	895	900	906	844	869	818	833	853	944	875	947	892		
		Edh at Noon	135	151	156	186	157	183	171	162	114	142	105	127		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	4.93	5.76	6.46	6.63	6.12	5.58	5.57	5.56	4.96	5.13	5.00	4.86		
		RadStd	0.32	0.48	0.37	0.25	0.39	0.40	0.47	0.31	0.36	0.37	0.38	0.27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a) Station Only	(b) Regional (7 neighbors)	(d) N/A	(e) +0.70	(f) -1.86	(g) N/A	(h) -0.94	(i) -1.31	(j) -30	(k) N/A	(l) N/A	(m) N/A
(a) Station Only	(b) Regional (7 neighbors)	(d) +0.37	(e) +1.05	(f) -1.75	(g) N/A	(h) N/A	(i) N/A	(j) -2	(k) -127	(l) +165	(m) +3

Nomenclature: See separate page